

Metaheuristic Algorithms and Applications: Program Report III

Program Report on Traveling Salesman Problem

Deadline: June 15, 2026

Apply the metaheuristic algorithms (Ant Colony Optimization, GA, PSO, etc.) to the Traveling Salesman Problem on **Traveling Salesman Problem (TSPLIB Dataset) provided by ZIYA**, which is referred to <https://www.kaggle.com/datasets/ziya07/traveling-salesman-problem-tsplib-dataset?resource=download>, (**tsp_dataset.csv**(225.98 MB)), and one can download this dataset.

About Dataset

This dataset is designed for solving the Traveling Salesman Problem (TSP). It includes 2,783 instances, where each instance represents a TSP problem with 20 cities, randomly distributed in a 2D space. The dataset facilitates benchmarking different optimization approaches by providing city coordinates, pairwise distance matrices, and categorized best routes as the target variable.

Dataset Features:

Instance_ID: Unique identifier for each TSP instance.

City Coordinates (X, Y): Randomly generated locations for 52 cities, the instance_id 24.

Total_Distance: Sum of pairwise distances (placeholder for optimization results).

Best_Route (Categorical Target): The best route category assigned for each instance, encoded as integers.

Use Cases:

Benchmarking metaheuristic algorithms. Evaluating local search enhancements. Training machine learning models for TSP route prediction. Testing optimization strategies on combinatorial problems. This dataset provides a structured approach to testing hybrid TSP optimization techniques, making it ideal for researchers, students, and practitioners in optimization and machine learning.

The Program Report should include the following two parts.

Part I: Each team should execute at least two different metaheuristic algorithms and make the comparisons by Best (shortest) Distance, Best fitness value, Average Distance, and Computation Time. Three cases are examined: (i) the first 12 cities, the first 30 cities, and (iii) all 52 cities.

Part II: Only the first 30 cities are considered; determine the shortest route path if all the even-numbered cities are not connected. Note that the fitness function is different from that in Part I.

Discussions:

- (i) Compare all the parameters you used, for example, please verify the Encoding of solutions (e.g., permutations), the Fitness function (total path length), the Selection, the crossover, the mutation strategy, and the stopping condition for the Genetic Algorithm. So do ACO, PSO, etc.
- (II) Path visualization: plot a map of the Best Roundtrip route for Parts I and II.

instance_id	num_cities
24	52

city_coordinates

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distance_matrix

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